# Maria Rosaria Briglia

*PhD student in AI for Government and Public Security, Sapienza University of Rome*
*Visiting researcher at NSF AI Institute for Foundations of Machine Learning (IFML), UT Austin with Prof. Adam Klivans — November 2025 – present*

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## Research

**Adversarial training through an energy lens**
Reinterpreting robust classifiers as energy-based models, which reframes catastrophic overfitting and robust overfitting as behaviour of the energy landscape rather than isolated training pathologies — and turns that reading into concrete regularizers and sample weighting schemes.

**The geometry of representation space**
Concept control and prompt safety depend on the shape of the space you are steering in. Euclidean adjustments to text embeddings are the standard tool; hyperbolic space, with parallel transport in place of vector arithmetic, gives more expressive and more stable manipulation of concepts.

**Threats to generative media, and defenses that hold**
Proactive image protection rests on an assumed threat model, and I test those assumptions — whether watermark-style defenses survive diffusion-based reconstruction that re-synthesises an image while preserving how it looks. The broader question is what a defense is actually promising once the attacker has generative models of their own.

**Generative models for proteins**
On the structure side, benchmarking antibody–antigen complex prediction at scale: cofolding thousands of complexes to ask not only whether models recover the right interface, but whether their own confidence scores can tell which prediction did. On the design side, decoding-time methods for protein engineering that leave the base model untouched. A model that ranks its own outputs badly is the same problem whether the output is an image or a fold.

## Publications

2026 — **Neutralizing Proactive Defense using Diffusion-based Upsampling**
*ACM Workshop on Information Hiding and Multimedia Security (IH&MMSec;)*

2026 — **Harnessing Hyperbolic Geometry for Harmful Prompt Detection and Sanitization**
*ICLR 2026*

2026 — **Semantic Steering via Hyperbolic Geometry**
*Beyond Euclidean Workshop, ECCV 2026 — Oral*

2026 — **Folding scFv–Antigen Complexes at Scale**
*GEM Workshop, ICLR 2026*

2025 — **What is Adversarial Training for Diffusion Models?**
*ICLR 2026*

2025 — **Implicit Inversion turns CLIP into a Decoder**
*ICLR 2026*

2025 — **Understanding Adversarial Training with Energy-based Models**
*arXiv preprint*

2024 — **Shedding More Light on Robust Classifiers under the Lens of Energy-based Models**
*ECCV 2024*

## Service

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| 2026 | Co-organizer, 3rd Workshop and Challenge on Unlearning and Model Editing (U&ME;), ECCV 2026 |
| 2026 | Member, The Good AI Lab |

## Education and positions

2025 Nov – present **Visiting researcher**
*NSF AI Institute for Foundations of Machine Learning (IFML), UT Austin*
Visiting period under the supervision of Prof. Adam Klivans, applying representation learning and mechanistic interpretability to protein engineering.

2026 – present **Member**
*The Good AI Lab*
A collective of researchers and engineers from universities, labs and companies, working on AI research with social impact.

2023 – present **PhD, National PhD programme in AI for Government and Public Security**
*Sapienza University of Rome*
Developing practical defense techniques against the media manipulation enabled by generative AI.

2021 – 2023 **M.Sc. Computer Engineering (LM-32), Artificial Intelligence and Intelligent Robotics**
*University of Salerno*
Thesis: Automatic Generation of Identity-Preserving Gesture Videos.

2023 **Research scholarship, artificial vision for autonomous driving on rail**
*RFI and University of Salerno*
Designed the on-board hardware acquisition system, and the real-time anomaly detection along the railway track.

2018 – 2021 **B.Sc. Computer Engineering (L-8)**
*University of Salerno*
Graduated 110/110 cum laude, and recognised for completing the degree with highest honours in the shortest time possible. Thesis: Automatic Speaker Recognition Using a Neural Network Integrated into a Web Portal.

## Summer schools

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| 2025 | **CISPA – ELLIS Summer School, Saarbrücken, Germany** |
| 2024 | **Generative Modeling Summer School, Eindhoven University of Technology** |

## Awards

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| 2023 | Recognition for an outstanding academic career, University of Salerno — for completing the B.Sc. with highest honours in the shortest time possible. |
| 2023 | Selected for courses reserved for excellent students, University of Salerno — open only to students averaging at least 28. |

## Technical

Python, C, Java, R, Matlab · PyTorch, TensorFlow · ROS Noetic, Django · Linux
Italian (native) · English (C1, Cambridge Advanced) · French (beginner)